Exercise A. Let $f(x) = \dfrac{x}{x^2+1}$.

A1

(a) Find the intervals where f is increasing or decreasing. Draw a number line to depict your answer, and record your final answer using interval notation.

$$f'(x) = \frac{(x^2+1)\cdot 1 - x(2x)}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}$$

$$f'(x) = 0 \Rightarrow 1-x^2=0 \Rightarrow (1-x)(1+x)=0 \Rightarrow x=1 \text{ or } -1$$

$$f'(x) \text{ DNE} \Rightarrow (x^2+1)^2 = 0 \Rightarrow \text{no such } x\,!$$

$-$ 0 $+$ 0 $-$

f'

-1 1

$$f'(-2) = \frac{1-(-2)^2}{((-2)^2+1)^2} = \frac{-3}{25} \ \ominus$$

$$f'(0) = \frac{1-0^2}{(0^2+1)^2} = 1 \ \oplus$$

$$f'(2) = \frac{1-2^2}{(2^2+1)^2} = \frac{-3}{25} \ \ominus$$

f is increasing for x in $(-1,1)$

f is decreasing for x in $(-\infty,-1)$ & $(1,\infty)$

A2

(b) Find the local maximum & local minimum points of f. Explicitly cite a relevant test in your response.

Both 1 & -1 are critical numbers of f b/c f' of both is 0 & both are in the domain of f. Using the 1st Deriv Test below, we have:

- f' changes signs from $-$ to $+$ at $x=-1 \Rightarrow (-1, f(-1))$ is a local min point of f
- f' changes signs from $+$ to $-$ at $x=1 \Rightarrow (1, f(1))$ is a local max point of f

Note: The next problem will test your conceptual understanding of what it means for a function to increase or decrease, as well as of the definition of a critical number. A variation of this problem appeared on an exam during the Fall 2023 semester.

A1

Exercise B. What would the critical numbers of the function f be if...

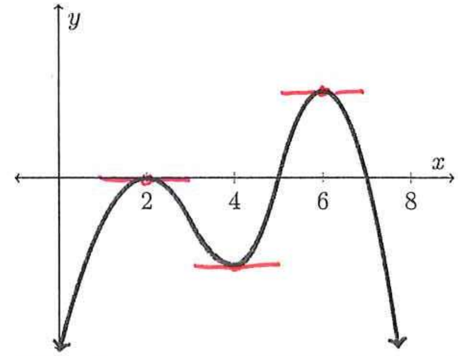
(a) the curve pictured to the right is the graph of f?

(b) the curve pictured to the right is the graph of f'?

On which intervals would the function f be *decreasing* if...

(c) the curve pictured to the right is the graph of f?

(d) the curve pictured to the right is the graph of f'?



ANSWER (a):	ANSWER (b):	ANSWER (c):	ANSWER (d):
$x=2,4,6$	$x=2,5,7$	$(2,4)$ & $(6,\infty)$	$(-\infty,2),(2,5),$ & $(7,\infty)$

Space is provided below for you to explain your answers.

(a) f' is zero at 2, 4, & 6 b/c the tangent lines are horizontal there $\Rightarrow$ 2, 4, & 6 are crit #s

(b) the height of the graph is zero at 2, 5, & 7 $\Rightarrow f'$ is 0 at 2, 5, & 7 $\Rightarrow$ 2, 5, & 7 are crit #s

(c) height decreases for x in $(2,4)$ & $(6,\infty)$

(d) height is negative for x in $(-\infty,2),(2,5)$, & $(7,\infty)$ $\Rightarrow$ f' is negative & thus f is decreasing on those intervals

Note: The next problem will test your comprehension of warm-up exercise 1 and the Chain Rule.

A1

Exam Practice 2. Find all critical numbers of $f(x)=\sqrt[3]{x^2-4x}$.

$$f(x)=(x^2-4x)^{1/3} \Rightarrow f'(x)=\frac{1}{3}(x^2-4x)^{-2/3}\cdot(2x-4)=\frac{2x-4}{3(x^2-4x)^{2/3}}$$

$$f'(x)=0 \Rightarrow 2x-4=0 \Rightarrow 2x=4 \Rightarrow x=2$$

$$f'(x) \text{ DNE} \Rightarrow 3(x^2-4x)^{2/3}=0 \Rightarrow (x^2-4x)^{2/3}=0 \Rightarrow x^2-4x=0 \Rightarrow x(x-4)=0$$

$$\Rightarrow x=0 \text{ \& } x=4$$

f has a domain of all real numbers $\Rightarrow f$ is continuous at $x=0,2,4$

$\Rightarrow \boxed{x=0,2,4}$ are critical numbers of f

Note: The next 2 exercises make use of all the information in this worksheet and the associated pre-class assignment. The goal is for you to continue to fluently apply the definition of a critical number, to find & interpret the sign of the first derivative as it relates to whether a function is increasing or decreasing, and to use the First Derivative Test to classify critical numbers.

Exam Practice 3. Let $f(x) = \ln(x^2 - 2x + 1)$.

A1

(a) Find all critical numbers of f.

$$f'(x) = \frac{1}{x^2-2x+1} \cdot (2x-2) = \frac{2(x-1)}{(x-1)^2} = \frac{2}{(x-1)}$$

$f'(x) = 0 \Rightarrow 2 = 0 \Rightarrow$ no such x

$f'(x)$ DNE $\Rightarrow x-1=0 \Rightarrow x=1$

$f(1) = \ln(1^2 - 2\cdot 1 + 1) = \ln(0)$, undefined $\Rightarrow x=0$ NOT a critical number

$\boxed{f \text{ has NO critical numbers}}$

4

A1

(b) Find the intervals on which f is increasing & the intervals on which f is decreasing.

f': $-$ on the left of 1, DNE at 1, $+$ on the right of 1

$$f'(-2) = \frac{2}{-2-1} = -\frac{2}{3} \ (-)$$

$$f'(2) = \frac{2}{2-1} = 2 \ (+)$$

$\boxed{\begin{array}{l}\text{increasing on: } (1, \infty) \\ \text{decreasing on: } (-\infty, 1)\end{array}}$

A2

(c) At which x-values does f have local maximum values? At which x-values does f have local minimum values? Explain.

f has $\boxed{\text{no local extreme values}}$! Although the sign of f' changes from $-$ to $+$ at $x=1$, $f(1)$ is undefined, so f cannot have a local extremum there. There is no other sign change in f', so f has no local extrema.

Exam Practice 4. Let $f(x) = \frac{1}{5}x^5 - \frac{5}{3}x^3 + 4x$.

A1

(a) Find the critical numbers of f.

$f'(x) = x^4 - 5x^2 + 4 = (x^2-4)(x^2-1) = (x-2)(x+2)(x-1)(x+1)$

$f'(x) = 0 \Rightarrow x = 2, -2, 1, -1$; $f'(x)$ DNE $\Rightarrow$ no such x!

f is continuous for all real numbers since its a polynomial so $\boxed{x = 2, -2, 1, -1}$ are critical numbers of f.

A1

(b) Find the intervals where f is increasing or decreasing.

$+ \; 0 \; - \; 0 \; + \; 0 \; - \; 0 \; +$ $\quad f'$

$-2 \quad -1 \quad 1 \quad 2$

x	(x−2)	(x−1)	(x+1)	(x+2)	$f'(x)$
−3	−	−	−	−	(−)(−)(−)(−) = (+)
−1.5	−	−	−	+	(−)(−)(−)(+) = (−)
0	−	−	+	+	(−)(−)(+)(+) = (+)
1.5	−	+	+	+	(−)(+)(+)(+) = (−)
3	+	+	+	+	(+)(+)(+)(+) = (+)

increasing: $(-\infty, -2)$, $(-1, 1)$, $(2, \infty)$

decreasing: $(-2, -1)$, $(1, 2)$

A2

(c) At which critical number(s) does f attain local extrema? Explain.

f attains local maxima at $x = -2$ & $x = 1$ as both are critical numbers at which f' changes from + to −.

f attains local minima at $x = -1$ & $x = 2$ as both are critical numbers at which f' changes from − to +.

(Conclusions are drawn using the 1st Derivative Test.)